

Literature status of the infinite-dimensional direction from arXiv:2311.09207

Scoped literature-identification packet

11 August 2026

Status: literature-implied answer to the broad infinite-dimensional direction. The later paper does not separately restate the source’s exact quasi-Metropolis time-domain formula, so that formula-specific scope is not claimed here.

1 Source direction

Chen, Kastoryano, and Gilyén construct an efficient exactly detailed-balanced noncommutative Gibbs sampler in finite dimension [1]. After Proposition B.2, the source states on PDF page 32 that its exact coherent term appears to persist in infinite dimension and leaves verification of the construction and analysis for future work.

The mathematically meaningful extension must handle an unbounded Hamiltonian: on an infinite-dimensional Hilbert space a bounded Hamiltonian cannot have a normal Gibbs density, because $e^{-\beta H}$ is bounded below by a positive multiple of the identity and therefore is not trace class.

2 Later result and identification

Becker, Rouzé, and Salzmänn explicitly cite the source as the first exact Lindbladian Gibbs sampler and then develop a rigorous infinite-dimensional theory for unbounded Hamiltonians on separable Hilbert spaces [2]. Their abstract and introduction state that they construct KMS-symmetric quantum Markov semigroups that are well posed and implementable on qubit hardware.

More precisely, Proposition 2.8 identifies a trace-class generator under explicit compatibility and boundedness hypotheses. Section 4.1 introduces the Gaussian-convoluted generator

$$\mathcal{L}_{\sigma_E, \hat{f}, H}$$

with the same characteristic coherent coefficient

$$\frac{i}{2} \tanh\left(\frac{\beta(\nu_1 - \nu_2)}{4}\right) (A_{\nu_1})^* A_{\nu_2},$$

and Proposition 4.3 proves that its closed form generates a completely Markov KMS-symmetric semigroup. The paper further proves controlled finite-rank approximations and circuit implementations, including singular Metropolis-type filters.

This supplies the rigorous generation, exact KMS symmetry, and truncation analysis requested by the broad future-work direction. The identification is classified as *literature implied*, rather than *literature already answered*, because the later authors do not advertise their theorem as a line-by-line verification of Proposition B.2 in arXiv:2311.09207.

3 Scope limitation

The later construction uses an equivalent subsequent sampler framework and Dirichlet-form generation theory. This packet therefore does not assert that arXiv:2604.01192 proves the validity of the exact quasi-Metropolis time-domain formula in Proposition B.2 under every conceivable domain hypothesis. It does establish that the broad infinite-dimensional exact-detailed-balance direction is no longer an open new-result target.

4 Search evidence

The bounded search through 11 August 2026 covered the run indexes; exact phrases from the source’s future-work sentence; citations to arXiv:2311.09207; the finite-dimensional follow-up arXiv:2404.05998; and the 2026 infinite-dimensional papers arXiv:2604.01192, 2604.06077, and 2604.15263. The decisive source is arXiv:2604.01192; arXiv:2604.06077 explicitly invokes it as the companion infinite-dimensional framework.

References

- [1] C.-F. Chen, M. J. Kastoryano, and A. Gilyén, *An efficient and exact noncommutative quantum Gibbs sampler*, arXiv:2311.09207. <https://arxiv.org/abs/2311.09207>.
- [2] S. Becker, C. Rouzé, and R. Salzmänn, *Quantum Gibbs Sampling in Infinite Dimensions: Generation, Mixing Times and Circuit Implementation*, arXiv:2604.01192 (2026). <https://arxiv.org/abs/2604.01192>.